B28: SQL SuperHero Program : Session 37

1 message

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How many triggers can we create on a table?

This question is very popular. . Always answer like:
1. We can create total 13 types of trigger (You all know what these 13 types, we explained in class and previous notes both)
2. but with the number of trigger, we can create as many as we want. There is no such restriction about number of triggers.

VIEWS

Views are a logical virtual table created by "select query" but the result is not stored anywhere in the disk and every time we need to fire the query when we need data,
so we always get updated or latest data from original tables.

The performance of the view depends on our select query. If we want to improve the performance of view we should avoid using join statements in our query or if we need multiple joins between tables always try to use the index-based column for joining as we know index-based columns are faster than a non-index-based column

VIEW, in essence, is a virtual table that does not physically exist. Rather, it is created by a query joining one or more tables.

Syntax

The syntax for the CREATE VIEW Statement is:

```
CREATE VIEW view_name AS
  SELECT columns
  FROM tables
  [WHERE conditions];
```

```
create view vishal
as select first_name, email, job_id
from hr.employees;


select * from vishal
```

FIRST_NAME	EMAIL	JOB_ID
Steven	SKING	AD_PRES
Neena	NKOCHHAR	AD_VP
Lex	LDEHAAN	AD_VP
Alexander	AHUNOLD	IT_PROG
Bruce	BERNST	IT_PROG
David	DAUSTIN	IT_PROG
Valli	VPATABAL	IT_PROG
[illegible]	[illegible]	[illegible]

Drop VIEW
Once an Oracle VIEW has been created, you can drop it with the Oracle DROP VIEW Statement.

Syntax
The syntax for the DROP VIEW Statement in Oracle/PLSQL is:

DROP VIEW view_name;
drop view vishal;

Question: Can you update the data in an Oracle VIEW?

Answer: A VIEW in Oracle is created by joining one or more tables. When you update record(s) in a VIEW, it updates the records in the underlying tables that make up the View.

So, yes, you can update the data in an Oracle VIEW providing you have the proper privileges to the underlying Oracle tables.

```
CREATE VIEW ANIRUDH AS SELECT NAME, SALARY FROM TEST
```

```
View created.
```

→ View created

```
SELECT * FROM ANIRUDH
```

Basically runs the actual query.

NAME	SALARY
VISHAL	1000
JAISWAL	1500

```
2 rows selected.
```

```
DELETE FROM ANIRUDH WHERE NAME = 'VISHAL'
```

```
1 row(s) deleted.
```

→ Deleted from View

```
SELECT * FROM TEST
```

ID	NAME	SALARY
2	JAISWAL	1500

→ Actually deleted from table

Question: can view take space in database?
Answer: No

Question: Does the Oracle View exist if the table is dropped from the database?

Answer: Yes, in Oracle, the VIEW continues to exist even after one of the tables (that the Oracle VIEW is based on) is dropped from the database. However, if you try to query the Oracle VIEW after the table has been dropped, you will receive a message indicating that the Oracle VIEW has errors.

If you recreate the table (the table that you had dropped), the Oracle VIEW will again be fine.

```
drop table employee

Table dropped.
```

→ Main table dropped

```
select * from vishal

ORA-04063: view "SQL_MQNNNGKIXWRTDFPVBCKYRJEON.VISHAL" has errors
```

More Details: https://docs.oracle.com/error-help/db/ora-04063

→ View has error

(But view exists)

When to use the Oracle view (Very Important explain with below example)
You can use views in many cases for different purposes. The most common uses of views are as follows:

Simplifying data retrieval.
Implementing data security.

--> **suppose in a company, we have an employee table that has records of emp name, address and salary. we have to give access to all employee data to the admin team.**
It is not a good idea that the admin team can see the salary of employees. so we will create a view from that employee table and that view contains only name and address,
which is useful for the admin team. Here we implement data security and hide salary data from the admin team,
by creating a view on the same table.

WITH CHECK OPTION

The WITH CHECK OPTION clause is used for an updatable view to prohibit the changes to the view that would produce rows which are not included in the defining query.

select * from employees where department_id = 90
-- 3 rows
create view test_normal_view as select * from employees where department_id = 90;

select * from test_normal_view;

insert into test_normal_view values (999,'vishal','jaiswal','vishau','515.123.4567','20-JAN-2021','VP',25000,null,100,

90);
-- 1 row inserted

select * from employees where department_id = 90
-- 4 rows
-- (We inserted row in view and it actually inserted record in main table)
update test_normal_view set salary = 24500 where department_id = 90 and employee_id = 999;
-- 1 row updated
select * from employees where department_id = 90;
-- we can see changes
-- (We updated row in view and it actually updated record in main table)

create view test_check_view as select * from employees where department_id = 90 with check option;

select * from test_check_view;
-- 4 rows

insert into test_check_view values (998,'vishal1','jaiswal1','vishau1','514.123.4567','21-JAN-2021','AVP',25000,null, 100,90);
-- allowed
select * from test_check_view;
-- 5 rows
-- we are able to insert, because inserted for dept_id = 90
insert into test_check_view values (998,'vishal1','jaiswal1','vishau1','514.123.4567','21-JAN-2021','AVP',25000,null, 100,100);
-- not allowed because insert rows for dept_id = 100 and when we created the view we created with check option on department_id = 90

- WITH READ ONLY Specify WITH READ ONLY to indicate that the table or view cannot be updated.
- WITH CHECK OPTION Specify WITH CHECK OPTION to indicate that Oracle Database prohibits any changes to the table or view that would produce rows that are not included in the subquery.

```
create view test2 as select * from regions with read only
```

NO one can do insert/update/delete

```
View created.
```

```
insert into test2 values (7, 'twttw')
```

```
ORA-42399: cannot perform a DML operation on a read-only view
```

More Details: https://docs.oracle.com/error-help/db/ora-42399

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Force View:

A view can be created even if the defining query of the view cannot be executed. We call such a view as view with errors.

For example, if a view refers to a non-existent table or an invalid column of an existing table or if the owner of the view does not have the required privileges, then the view can still be created and entered into the data dictionary.

We can create such views (i.e. view with errors) by using the FORCE option in the CREATE VIEW command. When Force command is used in the view syntax then even if select statement is invalid, VIEW gets created successfully.
The advantage of force view is that in future if view script becomes valid, we can start using the view.

How can I see all views in Oracle?

USER_VIEWS/DBA_VIEWS

```
create force view wsx as
select * from hr.wertyy
```

```
Errors: VIEW WSX
Line/Col: 0/0 ORA-00942: table or view does not exist

Line/Col: 0/0 ORA-54039: table must have at least one column that is not invisible
```

```
select * from wsx
```

created but have errors.

```
ORA-04063: view "SQL_KEOWUISATSJYVMPAHMKWLKOFH.WSX" has errors
```

```
select object_name, status from user_objects where object_type = 'VIEW'
```

Data Dictionary

OBJECT_NAME	STATUS
ABCD	VALID
QAZ	VALID
TEST2	VALID
TEST_CHECK_VIEW	VALID
TEST_NORMAL_VIEW	VALID
WSX	INVALID
XYZ	VALID

our view (forceView) in INVALID state.

7 rows selected.

Force views are created in real time environment as per the business needs because of two specific reasons

1. If the structure of the base table is not completely known (ie) in some development environments the base tables may not be created w/o knowing the complete structure and definitions.
2. Suppose in Agile environment, our table is esign in njext sprint, but in the current sprint, we have to use a view, so we can opt for that force view option.

Can we Create Primary Key on Views?
views are logical, view is just a name of the query. If the view itself does not exist in the database system physically then how can we create index or any constraint. So answer in NO